The Gradient–Flow Hinge

A spatial difference creates a gradient; material or interaction laws determine the response

Working rule: First identify the quantity that varies across space. Then form a difference per unit distance. Finally supply the physical closure that converts that gradient into a flow, field, or force.

Name: ____________________

Class: ____________________

Hinge worksheet series

1. Heat conduction: temperature gradient

For a uniform slab in steady conduction,

$$P_{\text{th}} = kA\frac{T_{\text{hot}} - T_{\text{cold}}}{L}.$$

A glass pane has $k = 0.80\,\text{W}\,\text{m}^{-1}\text{K}^{-1}$, area $1.5\,\text{m}^2$, thickness $5.0\,\text{mm}$ and temperature difference $18\,\text{K}$.

A. Represent the spatial difference

Sketch the pane in cross-section. Label the hot side, cold side, thickness L, area A and energy-transfer direction. Mark the temperature drop across the pane.

B. Calculate the flow rate

Calculate the thermal power transferred through the pane. Include the unit conversion for L and state the direction of net transfer.

C. Interrogate the proportionality

Predict the effect on P_{th} of separately doubling A, L and ΔT. Explain why ΔT alone does not determine the rate.

D. Identify the closure

What does k represent physically? State two assumptions required by the slab model.

2. Electric conduction: potential gradient

For an ohmic wire,

$$I = \frac{V}{R}, \qquad R = \rho\frac{L}{A}, \qquad I = \frac{A}{\rho}\frac{V}{L}.$$

A wire has $\rho = 1.7 \times 10^{-8}\,\Omega\,\text{m}$, length $2.0\,\text{m}$, area $0.50\,\text{mm}^2$ and potential difference $3.4\,\text{V}$.

A. Represent the conductor

Sketch the wire. Label its length, cross-sectional area, high-potential end, low-potential end and conventional-current direction.

B. Calculate resistance and current

Convert the area to m^2, then calculate R and I. Show each stage and include units.

C. Separate gradient, geometry and material

Explain the separate roles of V/L, A and ρ. Compare $1/\rho$ with the thermal conductivity k in case 1.

D. State the model boundary

What experimental evidence would indicate that the wire is no longer behaving ohmically? Why might heating matter?

3. Fields: potential gradient gives direction

For a uniform field,

$$E = -\frac{\Delta V}{\Delta x}, \qquad g = -\frac{\Delta V_g}{\Delta x}.$$

Two parallel plates are separated by $4.0\,\text{cm}$ and have a potential difference of $600\,\text{V}$. Use $e = 1.60 \times 10^{-19}\,\text{C}$.

A. Represent potential and field

Draw the plates and label the high- and low-potential sides. Add electric-field arrows and show where a proton is placed. Explain the minus sign geometrically.

B. Calculate field and force

Calculate the electric-field magnitude and the force on the proton. State the force direction.

C. Connect potential to energy

A proton is released from rest. State what happens to its electric potential energy and kinetic energy. Use $\Delta E_p = q\Delta V$ in your explanation.

D. Distinguish field from flow

Why is a field relation different from a conduction law? What additional relation is needed to predict the motion of the proton?

Synthesis: a shared gradient does not imply a shared mechanism

1. Complete the comparison.

Case	Spatial gradient	Response	Physical closure
Heat			
Charge			
Field			

2. Develop the central claim.
Respond to:

"A gradient indicates a directed tendency; it does not by itself determine the rate or response."

Use all three cases. Include the terms *difference*, *distance*, *direction*, *conductivity or resistivity*, *geometry*, *field*, and *closure*.

Teacher guide and indicative answers

The Gradient–Flow Hinge

1. Heat conduction

Representation. Energy is transferred from the hotter side to the colder side. The one-dimensional temperature gradient is approximated by $\Delta T/L$.

Calculation.

$$L = 5.0 \times 10^{-3}\ \mathrm{m},$$

$$P_{\mathrm{th}} = 0.80(1.5)\frac{18}{5.0 \times 10^{-3}} = 4.32 \times 10^{3}\ \mathrm{W}.$$

Scaling. Doubling A doubles the power; doubling L halves it; doubling ΔT doubles it.

Closure. Thermal conductivity k measures the material's capacity to conduct thermal energy. The simple relation assumes steady, approximately one-dimensional transfer, constant k, uniform area and negligible contact resistance.

Diagnostic distinction. Temperature is a state variable. Heat is energy transferred because a temperature difference exists.

2. Electric conduction

Conversion.

$$0.50\ \mathrm{mm}^2 = 5.0 \times 10^{-7}\ \mathrm{m}^2.$$

Calculation.

$$R = \rho\frac{L}{A} = \frac{(1.7 \times 10^{-8})(2.0)}{5.0 \times 10^{-7}} = 6.8 \times 10^{-2}\ \Omega,$$

$$I = \frac{3.4}{0.068} = 50\ \mathrm{A}.$$

Conventional current is directed through the resistor from higher to lower potential.

Roles. V/L supplies the spatial potential gradient, A provides the available cross-section, and ρ characterises material opposition to current. Electrical conductivity is $1/\rho$, analogous in role to k for thermal conduction.

Model boundary. Curvature in the I–V graph, or a changing resistance as current rises, indicates non-ohmic behaviour. Joule heating may change the resistivity.

3. Field from potential gradient

Field magnitude.

$$E = \frac{600}{0.040} = 1.5 \times 10^{4}\ \mathrm{N\,C^{-1}}.$$

Force on a proton.

$$F = qE = (1.60 \times 10^{-19})(1.5 \times 10^{4}) = 2.4 \times 10^{-15}\ \mathrm{N}.$$

The proton accelerates in the field direction, towards lower electric potential.

Energy. For motion with the field, $\Delta V < 0$ and

$$\Delta E_p = q\Delta V < 0.$$

Potential energy decreases while kinetic energy increases.

Minus sign. The field points in the direction in which potential decreases most rapidly. Potential itself is reference-dependent; its spatial change determines the field.

Field versus flow. The field relation defines force per unit charge. It is not a material flow law. Newton's second law, $F = ma$, is additionally required to predict the proton's motion.

Teacher synthesis: what should survive the calculations?

Indicative comparison.

Case	Gradient	Response	Closure
Heat	$\Delta T/L$	thermal power or heat flux	k, area and steady-state model
Charge	V/L	current or current density	ρ, area and ohmic regime
Field	$-\Delta V/\Delta x$	field strength and force	interaction law and coupling property

Core interpretation. All three cases begin with a quantity that changes across space. The ratio of change to distance supplies magnitude and direction information. The models then diverge. Heat and charge conduction require constitutive properties and geometry to determine a rate. A potential gradient defines a field; a separate dynamical law determines the resulting motion.

Expected conclusion. The gradient is the shared mathematical hinge. Conductivity, resistivity, geometry, interaction law and coupling property supply the physical closure.